CONFIDENTIAL · BUSINESS INTELLIGENCE REPORT

CUSTOMER INTELLIGENCE · MARKET SEGMENTATION

AEROFIT

TREADMILL INTELLIGENCE REPORT

Who Buys What — and Why:

A Probabilistic Study of Treadmill Buyer Behaviour

ANALYST	COURSE	DATE	DATASET
Devasish Mishra	Prob. & Statistics	March 2026	180 Customers

CONFIDENTIAL · BUSINESS INTELLIGENCE REPORT

CUSTOMER INTELLIGENCE · MARKET SEGMENTATION

AEROFIT

TREADMILL INTELLIGENCE REPORT

Who Buys What — and Why:

A Probabilistic Study of Treadmill Buyer Behaviour

ANALYST	COURSE	DATE	DATASET
Devasish Mishra	Prob. & Statistics	March 2026	180 Customers

SECTION 01

Executive Summary

Business context, objectives, and key findings at a glance

PROBLEM STATEMENT

"Aerofit sells three treadmill models — KP281, KP481, and KP781. The company lacks clarity on which customer profiles gravitate toward each product, limiting marketing efficiency and revenue potential."

This study analyses 180 verified customer purchase records across 9 variables — demographics, income, fitness level, and usage habits — to build a probabilistic understanding of Aerofit's buyer landscape and translate it into targeted business strategy.

DATASET SNAPSHOT

180	9	0	28.8	$53.7K	3
Customers	Variables	Missing Values	Avg. Age (yrs)	Avg. Income	Product Lines

KEY FINDINGS AT A GLANCE

Income is the #1 Predictor

100% of very high-income buyers ($80K+) chose KP781. Zero low-income buyers chose it. The income–product relationship is near-deterministic.

A Stark Gender Gap at Premium

Only 17.5% of KP781 buyers are female — despite a perfect 50/50 split at entry level. This represents significant untapped revenue.

KP281 Drives Volume, KP781 Drives Margin

KP281 commands 44% market share. KP781 — though only 22% — captures the highest-value customers with premium pricing power.

Fitness & Usage Amplify Income Signal

High fitness (4–5 rating) and frequent usage (5–6x/week) are strong secondary signals for KP781 candidacy, even without direct income data.

SECTION 02

Data Overview & Quality Audit

Structural inspection, completeness check, and statistical profile of the dataset

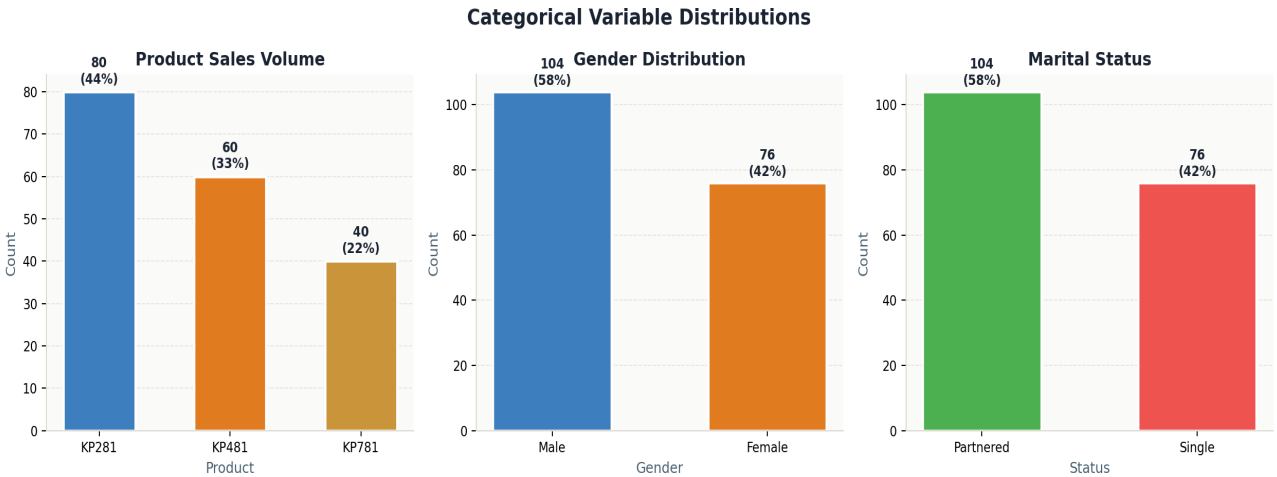

Figure 2.1 — Categorical Variable Distributions: Product, Gender, Marital Status

Dataset Quality

The Aerofit dataset is publication-quality — 180 complete records with zero nulls across all 9 columns. No imputation, dropping, or encoding required. High confidence in all statistical outputs.

Population Profile

Average customer: 28.8 yrs old, earns ~$53.7K/year, exercises 3.5x/week, and self-rates fitness at 3.3/5. This baseline masks dramatic variation across product tiers.

Product Mix Signal

KP281 (entry) commands 44.4% of purchases — nearly half the portfolio. This volume concentration may indicate under-promotion of mid and premium models.

Gender Balance

Overall 57.8% male, 42.2% female — near-balanced at aggregate level. However, this split diverges dramatically across product tiers, a central finding of this study.

SECTION 03

Exploratory Data Analysis

Uncovering patterns, distributions, and relationships across customer attributes

3.1 Age Distribution

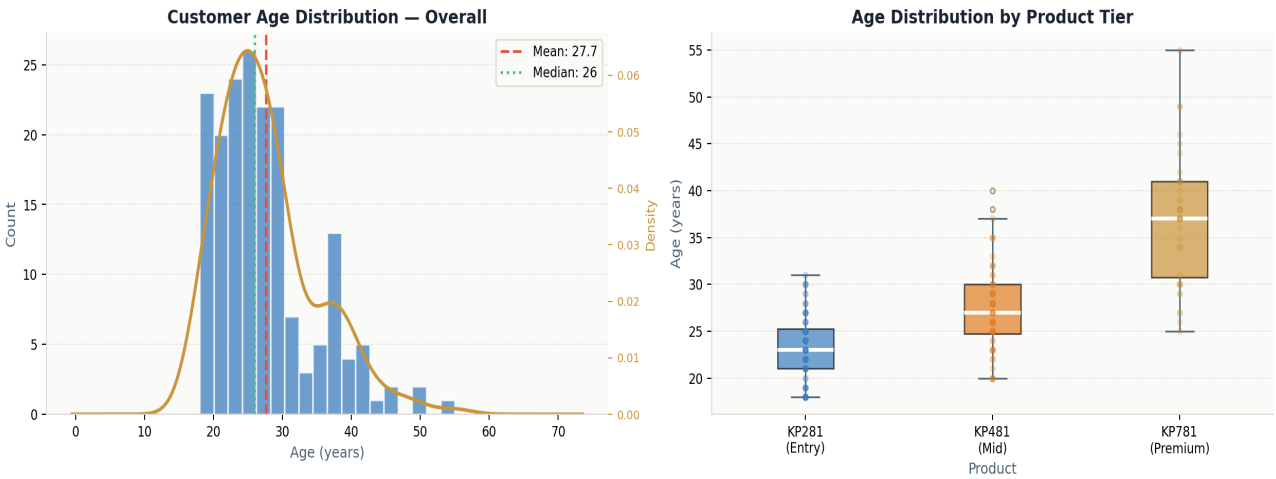

Figure 3.1 — Age distribution overall (left) and by product tier (right)

Young-Dominated Base

Distribution peaks at 20–28 years. Mean = 28.8, Median = 26 — confirming early-career young adults as Aerofit's primary segment.

Age Stratifies by Tier

KP281 buyers median ~24; KP481 ~26; KP781 skews toward 30–45 year-olds. Clear positive age–tier relationship.

Marketing Implication

KP281 → Instagram/YouTube (Gen Z). KP781 → LinkedIn/professional media (established professionals).

3.2 Income & Fitness by Product Tier

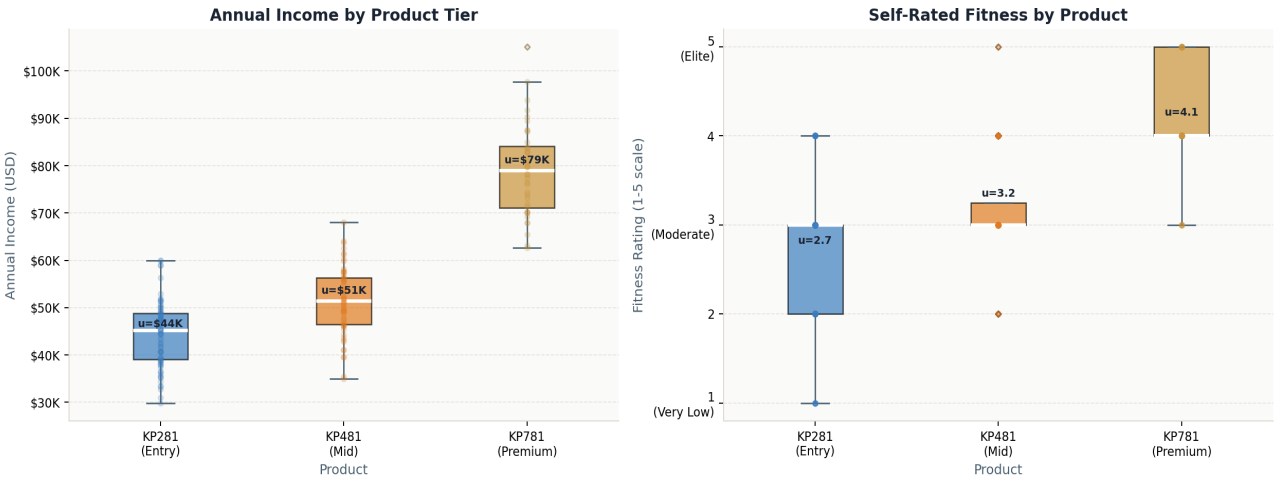

Figure 3.2 — Annual income (left) and self-rated fitness (right) by product tier

Income: The Clearest Differentiator

KP281 buyers avg ~$44K; KP481 ~$51K; KP781 ~$76K. Inter-quartile ranges barely overlap — near-perfect income segmentation with a clear, actionable gradient.

Fitness: A Powerful Secondary Signal

KP781 buyers self-rate 4–5; KP281 buyers cluster at 2–3. A buyer with income >$70K and fitness rating >= 4 is an extremely high-probability KP781 prospect.

3.3 Usage Frequency & Gender Split

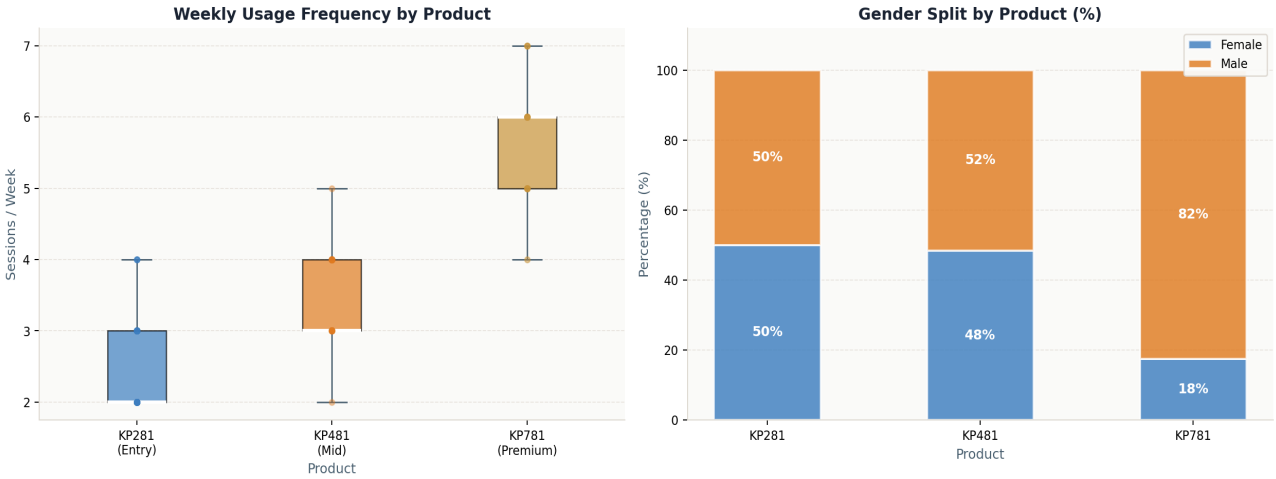

Figure 3.3 — Weekly usage by product (left) and gender split per product (right)

Usage: Commitment Scales with Product

KP781 owners average 5–6 sessions/week — nearly double KP281's 2–3 sessions. Premium buyers purchase with far greater intentionality and discipline.

Gender Gap Widens at Premium

KP281 is perfectly 50/50. KP781 is 82.5% male vs 17.5% female. This widening gap points to gaps in marketing reach and product perception — not preference.

3.4 Correlation Analysis

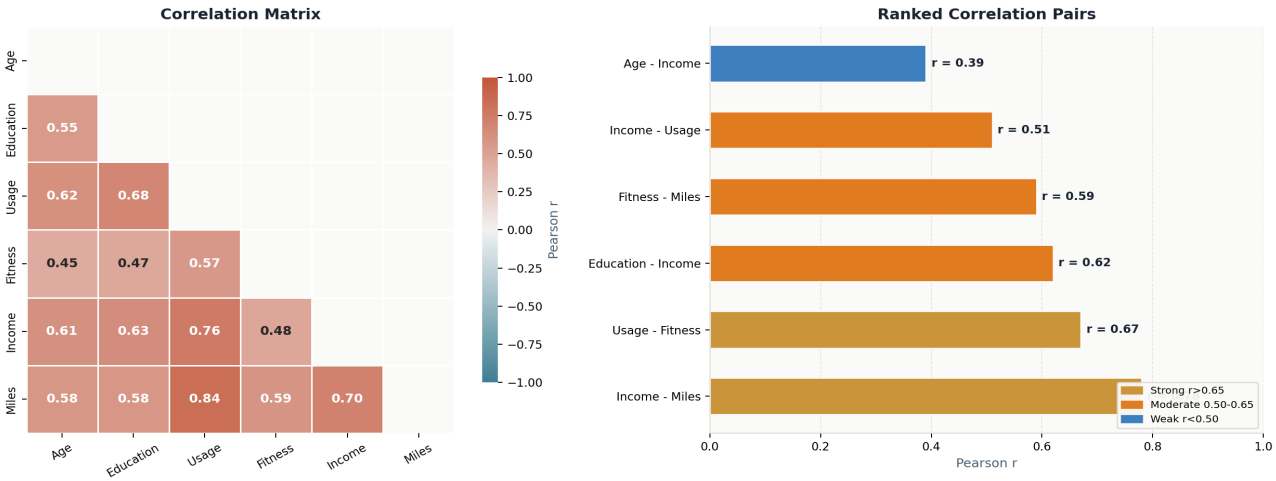

Figure 3.4 — Correlation heatmap (left) and ranked correlation pairs (right)

Income - Miles (r=0.78)

Strongest link. High earners don't just buy better treadmills — they use them more intensively, consistent with research on income and fitness habits.

Usage - Fitness (r=0.67)

Frequent users rate their fitness higher. These two dimensions largely describe the same customer — and they overwhelmingly prefer KP781.

Education - Income (r=0.62)

Education is a useful income proxy when direct data is unavailable, making it a viable alternative for ad targeting on Google and Meta.

SECTION 04

Probability Analysis

Marginal and conditional probabilities to quantify purchasing likelihood by segment

4.1 Marginal Probability P(Product)

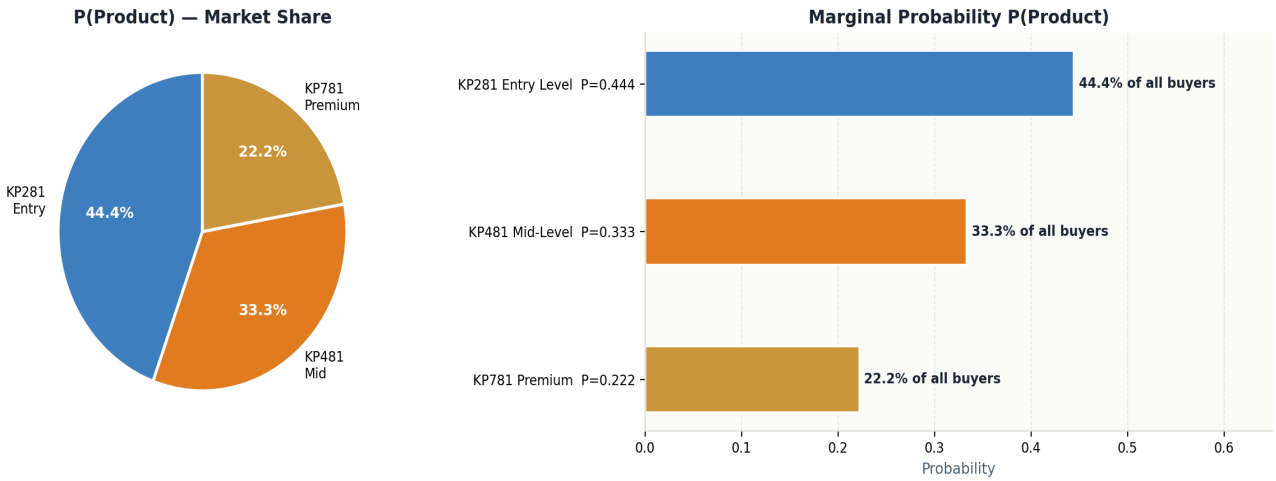

Figure 4.1 — Product market share pie chart (left) and marginal probability bar chart (right)

4.2 Conditional Probability P(Product | Gender)

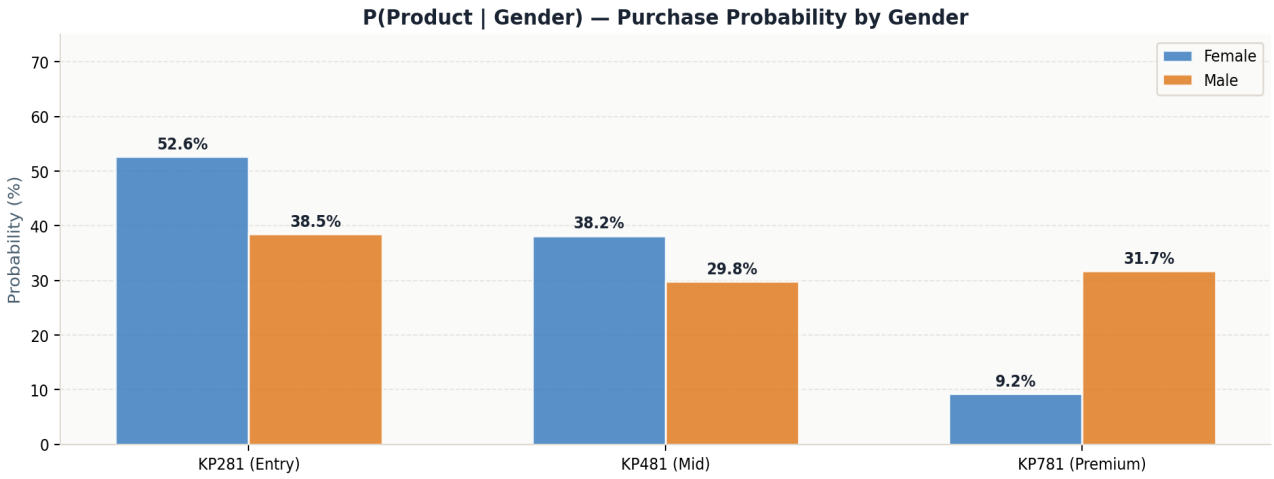

Figure 4.2 — Probability of choosing each product, broken down by gender

4.3 Conditional Probability P(Product | Income Group)

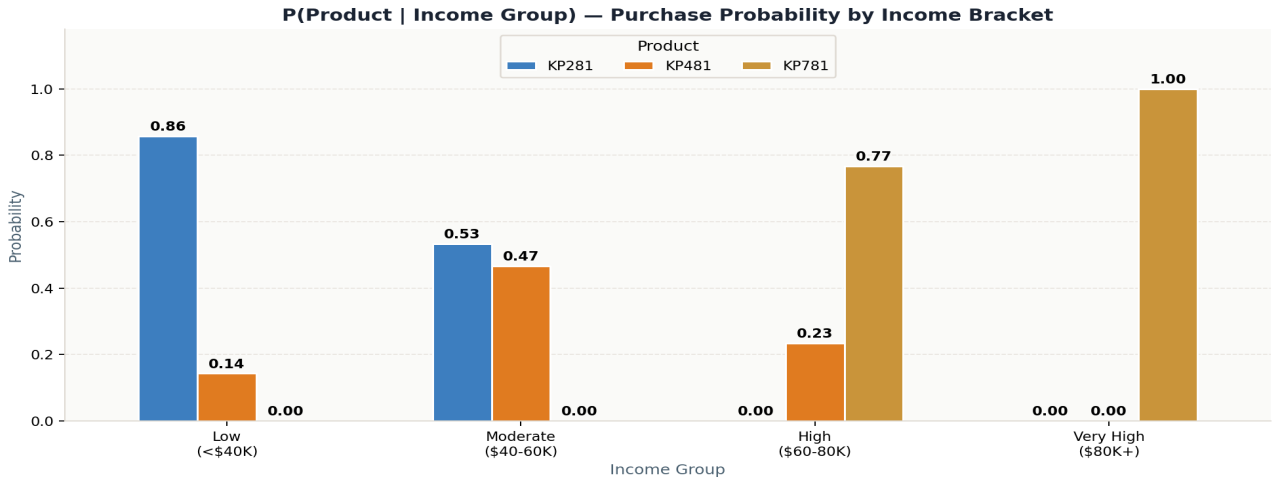

Figure 4.3 — Purchase probability by income bracket; note P(KP781 | Very High Income) = 1.00

P = 0.444

P(KP281) — Volume Leader

Nearly 1 in 2 Aerofit buyers chooses the entry model. Driven by a large base of young, budget-conscious customers.

P = 1.000

P(KP781 | Very High Income)

Every customer earning above $80K purchased KP781. This 100% hit rate makes income the most deterministic targeting variable in the dataset.

3.4x

Male vs Female KP781 Likelihood

P(KP781|Male) = 0.317 vs P(KP781|Female) = 0.092. Men 3.4x more likely. This is a marketing problem — not a preference problem.

SECTION 05

Customer Segmentation Profiles

Three distinct buyer archetypes — each requiring a fundamentally different marketing approach

KP281
ENTRY LEVEL

44.4%
MARKET SHARE

Age Range	**18 – 28 years**
Annual Income	**Below $40,000**
Fitness Level	**Beginner (2–3 / 5)**
Weekly Usage	**2–3 sessions**
Gender Split	**50% F / 50% M**
Decision Driver	**Price & accessibility**
Education	**~14 years**

Marketing Strategy: Instagram / YouTube campaigns · EMI financing · Student discounts · "Start your fitness journey" narrative

KP481
MID LEVEL

33.3%
MARKET SHARE

Age Range	**24 – 35 years**
Annual Income	**$40,000 – $60,000**
Fitness Level	**Moderate (3 / 5)**
Weekly Usage	**3–4 sessions**
Gender Split	**48% F / 52% M**
Decision Driver	**Features vs. price**
Education	**~16 years**

Marketing Strategy: Google / Facebook campaigns · Feature comparison content · "Take your fitness further" narrative

KP781
PREMIUM

22.2%
MARKET SHARE

Age Range	**28 – 50 years**
Annual Income	**Above $80,000**
Fitness Level	**Advanced (4–5 / 5)**
Weekly Usage	**5–6 sessions**
Gender Split	**18% F / 82% M**
Decision Driver	**Performance & specs**
Education	**~18 years**

Marketing Strategy: LinkedIn / Premium media · Performance data · ROI framing · "Built for champions" narrative

SECTION 06

Strategic Business Recommendations

Six data-backed actions to optimise marketing, increase revenue, and close the gender gap

KP281 GROWTH

01. Dominate the Entry Market

KP281's 50/50 gender balance is its strongest asset. Launch social-first campaigns on Instagram and YouTube targeting 18–28 yr-olds earning <$40K. Lead with "start your fitness journey." Offer EMI financing and student discounts to lower barriers. Goal: convert the 18–24 cohort before they graduate to competitors.

KP781 REVENUE

02. Target the $80K+ Segment Precisely

P(KP781 | Income >$80K) = 1.00 — this segment converts with 100% certainty. Invest in LinkedIn and premium digital advertising targeting high-earning professionals by job title (finance, consulting, tech). Focus on performance specs, time-efficiency, and ROI of fitness. Showroom demos and luxury partnerships amplify reach.

GENDER GAP

03. Close the Female Premium Gap

With only 17.5% of KP781 buyers female, there is significant revenue opportunity. Partner with female fitness athletes and influencers. Feature women prominently in KP781 campaigns. A dedicated "Women in Performance" campaign can materially shift this ratio — the data suggests it is a marketing gap, not a preference gap.

AD TARGETING

04. Income-Inferred Ad Segmentation

Implement income-proxied audiences on Google & Meta. Route <$40K segments to KP281 funnels; $40–60K to KP481; $60K+ directly to KP781 landing pages. Use job title and education level as income proxies. Expected outcome: significant ROAS improvement and reduction in wasted ad impressions.

WEBSITE CRO

05. Deploy a Fitness Onboarding Quiz

Build a 60-second "Find Your Treadmill" quiz on the Aerofit website. Map 3–5 questions (fitness level, weekly goals, budget range) to the most suitable product. Routes beginners to KP281; advanced high-intent users to KP781. Reduces decision friction and can meaningfully increase average order value.

RETENTION & UPSELL

06. Usage-Based Upgrade Programme

High-usage KP281/KP481 customers (5–6x/week) are the highest-probability KP781 upgraders. Build a CRM-based loyalty programme with milestone rewards: "You've run 500 miles — upgrade to KP781 with 15% off." Transforms your existing base into a premium revenue pipeline with minimal acquisition cost.

SECTION 07

Conclusion & Key Takeaways

Synthesising the evidence into a clear, actionable business verdict

> *"Customer purchasing behaviour at Aerofit is not random — it is structured, predictable, and — importantly — actionable. Income, fitness commitment, and demographic profile together explain the overwhelming majority of product choice."*
>
> — STUDY CONCLUSION, MARCH 2026

KEY TAKEAWAYS

01

Income is the single most reliable predictor of product choice.

Near-deterministic relationship. Every $80K+ earner chose KP781; every low-income buyer chose KP281/KP481. This makes income the cornerstone of any effective segmentation strategy.

02

KP281 dominates volume; KP781 dominates value.

44% market share for KP281. KP781's 22% represents the highest-income, highest-engagement, highest-margin cluster. Both tiers deserve dedicated investment — with entirely different marketing logic.

03

The female premium gap is a marketing problem, not a preference problem.

KP281 achieves 50/50 gender balance but KP781 is 82.5% male. This divergence is caused by how premium products are marketed — not inherent preference differences. Targeted campaigns can close this gap.

04

Fitness score and usage frequency are powerful qualifying signals.

A customer with fitness 4–5 and training 5+ times/week is almost certainly a KP781 prospect. Build these signals into website recommendation engines, CRM scoring, and sales conversations.

05

The data is clean, the patterns are strong, and the actions are clear.

180 records deliver statistically robust, low-ambiguity segments. Begin with income-targeted ad segmentation and the female premium campaign — the fastest paths to measurable ROI improvement.

Income = #1 Predictor | KP781 = High Margin | Female Gap = Opportunity | Fitness = Secondary Signal | Usage = Upsell Trigger